Exponential Speed-up of Quantum Cooling via Non-Local Quantum Expanders and Its Limitations

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Abstract

We explore an open quantum system architecture utilizing non-local quantum expander dynamics to achieve exponential improvements in mixing time during thermalization. By engineering energy-conserving jump operators along an expander graph, the system attains a size-independent Lindbladian spectral gap. However, we demonstrate that when the system Hamiltonian encodes a computationally hard problem, stringent thermodynamic and physical constraints offset this advantage. Specifically, high-precision ground state cooling requires an effective bath inverse temperature scaling as $\Omega(n \cdot 2^{\alpha n})$, translating to a super-exponential precision requirement on the bath reset-to-exchange rate ratio. Furthermore, compiling the necessary non-local jump operators on a two-dimensional superconducting architecture yields a total circuit depth of $\Omega(\sqrt{N} \log N)$. Thus, the practical advantage over diffusive cooling is sub-polynomial rather than exponential for now, though this is sensitive to future improvements.

Keywords: Quantum Thermodynamics, Quantum Chaos, Expander Graphs, Computational Complexity, Open Quantum Systems, Lindbladian Dynamics, Quantum Annealing

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I. INTRODUCTION

The initialization of quantum systems into their ground states is a fundamental prerequisite for various quantum technologies, most notably adiabatic quantum computing [1] and quantum annealing [2]. When a system Hamiltonian H_S encodes a computationally hard problem (for instance, an NP-hard Ising spin glass or a MAX-2-SAT instance), finding its ground state is intrinsically difficult. Traditional algorithmic cooling [3] or coupling to local, spatially constrained thermal baths results in slow, diffusive thermalization, often requiring time exponential in the system size to traverse the rugged energy landscape.

A critical measure is the *spectral gap* of the thermalization process. For a Lindblad master equation governing the joint system-bath density matrix, the spectral gap $\Delta_{\mathcal{L}}$ of the Lindbladian superoperator $\mathcal{L}$ sets the inverse of the characteristic relaxation time. In diffusive models based on nearest-neighbor interactions on a d-dimensional lattice, the spectral gap closes polynomially with system size, $\Delta_{\mathcal{L}} \sim \mathcal{O}(1/N^{2/d})$, so the mixing time $\tau_{mix} \sim \mathcal{O}(N^{2/d})$ is polynomial but can be extremely slow in practice. For one-dimensional spin chains, $\tau_{mix} \sim \mathcal{O}(N^2)$.

To bypass this slow local mixing, one can look to quantum chaos and quantum information theory, specifically the concept of *quantum expanders* [4, 5]. Expander graphs are sparse networks characterized by high connectivity and a uniform spectral gap in their adjacency spectra. A Lindbladian mapped to such a topology guarantees rapid, exponential mixing of quantum states, with $\tau_{mix} \sim \mathcal{O}(\log N)$ [6]. This point has surprisingly not been much discussed in the literature, despite its importance.

However, realizing a system-bath interaction as a quantum expander presents profound thermodynamic and computational challenges. Furthermore, as we will show, the complexity of solving a hard problem cannot be circumvented purely through engineered quantum expander dissipation, due to what we term the bath purity problem.

The paper is organized as follows. In Sec. II we review background on quantum expanders, Lindbladian mixing, and complexity-theoretic lower bounds on cooling. In Sec. III we formally construct the expander Lindbladian and establish its basic properties. In Sec. IV we carry out a complete spectral analysis. In Sec. V we state and prove the main bottleneck results. In Sec. VI we compare our architecture to standard diffusive cooling. In Sec. VII we discuss potential mitigations and fundamental limitations. We conclude in Sec. VIII.

II. BACKGROUND AND RELATED WORK

A. Expander Graphs and Their Spectral Properties

An undirected d-regular graph $G = (V, E)$ on $n = |V|$ vertices has adjacency matrix A with eigenvalues $d = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. The *spectral gap* is $\Delta_G = d - \lambda_2$. The graph is an *expander* if $\Delta_G = \Omega(d)$, i.e., the gap is a constant fraction of the degree. The celebrated Alon-Boppana theorem establishes a lower bound $\lambda_2 \geq 2\sqrt{d-1} - o(1)$; graphs saturating this bound, $\lambda_2 \leq 2\sqrt{d-1}$, are called *Ramanujan graphs* [7] and achieve the optimal expansion for their degree. Explicit constructions of Ramanujan graphs for any prime degree d exist via the Lubotzky-Phillips-Sarnak construction [7] as well as more recent algebraic methods [8].

The key property we exploit is the *mixing rate*: random walks on expander graphs mix in $\mathcal{O}(\log n)$ steps, where the convergence rate per step is governed by the ratio $\lambda_2/d < 1$. For Ramanujan graphs, this ratio approaches $2/\sqrt{d}$, which is optimal.

B. Quantum Expanders and Lindbladian Mixing

The quantum analogue of an expander is a quantum channel $\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ whose induced superoperator has a spectral gap bounded away from zero uniformly in the dimension [4]. Quantum expanders generalize the notion of classical spectral expansion to non-commutative probability spaces. Efficient constructions based on random unitary designs [9] and algebraic structures [5] are known.

In the Lindbladian context, the relevant object is the spectral gap $\Delta_{\mathcal{L}}$ of the Lindbladian superoperator. A series of works [10, 11] has established connections between the gap of the Lindbladian and rapid mixing of quantum states, deriving exponential convergence bounds analogous to classical Poincaré inequalities. In particular, for a primitive (i.e., full-rank steady state) Lindbladian, exponential convergence in trace norm holds:

$$\|\rho(t) - \rho_{ss}\|_1 \leq C \cdot e^{-\Delta_{\mathcal{L}} t}, \tag{1}$$

where C depends on the initial state and ρ_{ss} is the unique steady state; see [10] for the precise log-Sobolev derivation.

C. Complexity Constraints on Ground State Cooling

The question of whether engineered dissipation can efficiently prepare ground states of hard Hamiltonians is closely tied to complexity theory. Cubitt, Perez-Garcia, and Wolf [12] established that the spectral gap problem for local Hamiltonians is undecidable, implying severe limitations on any general cooling protocol. More directly relevant are results connecting the hardness of ground state preparation to QMA-completeness. Bravyi [13] introduced the quantum k-SAT problem and established QMA_1-hardness for $k \geq 4$; the QMA_1-completeness of quantum 3-SAT was subsequently proved by Gosset and Nagaj [14]. Aharonov, Gottesman, Irani, and Kempe [15] demonstrated QMA-hardness for systems on a line. Any cooling protocol that solves these problems efficiently would imply unexpected complexity-theoretic collapses.

These results motivate our investigation: even with exponentially fast *mixing*, the cooling protocol cannot escape the intrinsic hardness of the problem when that hardness manifests through the thermodynamic requirements on the environment. Our work translates this complexity-theoretic lower bound into a concrete physical statement about bath purity. Though once this bath purity problem is overcome, and other practical implementation difficulties are resolved, exponential improvement in mixing time is provided.

D. Algorithmic Cooling and Purity Limits

Algorithmic cooling [3] attempts to compress entropy from a subset of qubits into others, effectively cooling a target register at the expense of a "reset" register. The fundamental bound from thermodynamics, specifically the Landauer principle, states that erasing one bit of information requires dissipating at least $k_B T \ln 2$ of energy into the environment. In the quantum setting, the achievable purity after r rounds of compression is bounded by the initial purity of the reset qubits, establishing a bottleneck that is precisely what we find in the expander architecture. Our work shows that this purity bottleneck survives even when the mixing dynamics are exponentially accelerated by the expander topology.

III. MATHEMATICAL FRAMEWORK OF THE EXPANDER SETUP

A. Composite Hilbert Space and Hamiltonians

Consider a composite Hilbert space $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$, where $\mathcal{H}_S \cong (\mathbb{C}^2)^{\otimes n}$ is the n-qubit system to be cooled and $\mathcal{H}_B \cong (\mathbb{C}^2)^{\otimes N}$ is a bath composed of N identical qubits. We allow N to differ from n in general; in Definition 2 below we note the special case $N = n$ as the canonical balanced setting. The free Hamiltonian is:

$$H_0 = H_S \otimes I_B + I_S \otimes H_B, \tag{2}$$

where H_S is the system Hamiltonian encoding the problem of interest, and $H_B = \omega_0 \sum_{k=1}^{N} \sigma_z^{(k)}/2$ is a uniform transverse field on the bath qubits setting the characteristic energy scale ω_0.

The system Hamiltonian is taken to be a k-local Ising-type Hamiltonian:

$$H_S = -\sum_C J_C \prod_{i \in C} \sigma_z^{(i)}, \tag{3}$$

where the sum runs over clauses C of size at most k, and the couplings $J_C \in \mathbb{R}$ are chosen to encode a hard optimization instance. The ground state energy is E_0, the first excited state energy is E_1, and the energy gap is $\Delta E = E_1 - E_0$.

Definition 1 (Hard Hamiltonian)**.** *We say H_S is a* hard Hamiltonian *if the spectral gap satisfies $\Delta E \leq \Delta_0 \cdot 2^{-\alpha n}$ for some constants $\Delta_0, \alpha > 0$, i.e., the gap closes exponentially in the system size. This is generically expected for random instances of NP-hard problems in the worst case.*

B. The Interaction Graph and Jump Operators

To facilitate exponential cooling, we define an interaction graph $G = (V, E)$ where $V = [n] \cup [N]$ is the combined set of system and bath qubit indices, and E forms a d-regular bipartite expander graph with edges crossing exclusively between system qubits and bath qubits.

Definition 2 (Bipartite Expander Bath Graph)**.** *A bipartite d-regular graph $G = (S \cup B, E)$ with $|S| = n$ and $|B| = N$ is a (δ, d)-bipartite expander if for every subset $W \subseteq S$ with $|W| \leq n/2$, the neighborhood $\Gamma(W) \subseteq B$ satisfies $|\Gamma(W)| \geq (1 + \delta)|W|$. Equivalently, the*

singular value gap of the biadjacency matrix satisfies $\sigma_1 - \sigma_2 \geq \delta \cdot d$. The canonical case $N = n$ yields a balanced bipartite expander; we use this as the primary setting in Sections IV and V and note where results extend to general N.

For every edge $(i, j) \in E$ (with $i \in S$ a system qubit and $j \in B$ a bath qubit), we engineer an energy-conserving jump operator. The energy-conservation condition is $[H_0, L_{ij}] = 0$, which ensures that excitations are exchanged between system and bath without net energy injection. For resonantly matched qubits ($\omega_0 = \bar{E}$, where $\bar{E}$ is a representative system energy spacing), the jump operators take the form of non-local *excitation-exchange* (XX+YY) operators:

$$L_{ij} = \sqrt{\gamma} \left(\sigma_-^{(i)} \sigma_+^{(j)} + \sigma_+^{(i)} \sigma_-^{(j)} \right), \tag{4}$$

where $\gamma > 0$ is the coupling rate per edge and $\sigma_\pm = (\sigma_x \pm i\sigma_y)/2$ are the qubit raising and lowering operators. We remark that these simple two-body operators are used for the spectral analysis of Section IV; thermodynamically selective cooling in general requires energy-selective variants, a point we address in detail in Section V C. The full Lindblad master equation governing the composite density matrix $\rho \in \mathcal{D}(\mathcal{H})$ is:

$$\frac{d\rho}{dt} = \mathcal{L}[\rho] \equiv -i[H_0, \rho] + \sum_{(i,j)\in E} \mathcal{D}_{L_{ij}}[\rho], \tag{5}$$

where the Lindblad dissipator is:

$$\mathcal{D}_L[\rho] \equiv L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}. \tag{6}$$

The composite Lindbladian $\mathcal{L}$ is a superoperator on $\mathcal{B}(\mathcal{H})$. Since $\mathcal{H}$ has dimension 2^{n+N}, the superoperator acts on a space of dimension 4^{n+N}.

C. Continuous Bath Resupply

To prevent the system and bath from equilibrating at a shared finite excitation density, we couple the primary bath B to a secondary, ultra-cold Markovian reservoir at temperature $T_r \ll T_B$. This continuous resupply is modeled by local amplitude-damping dissipators acting exclusively on the bath nodes:

$$\mathcal{D}_{reset}[\rho] = \kappa \sum_{k=1}^{N} \left(\sigma_-^{(k)} \rho \sigma_+^{(k)} - \frac{1}{2}\{\sigma_+^{(k)} \sigma_-^{(k)}, \rho\} \right), \tag{7}$$

where $\kappa > 0$ is the reset rate. This dissipator drives each bath qubit individually toward the ground state $|0\rangle\langle 0|$ at rate κ, independent of the system. The full effective Lindbladian is thus:

$$\mathcal{L}_{full} = \mathcal{L} + \mathcal{L}_{reset}, \tag{8}$$

where $\mathcal{L}_{reset}[\rho] = \mathcal{D}_{reset}[\rho]$.

Remark 1. *The physical picture goes as: the system S exchanges excitations with the primary bath B via the expander jump operators, and the bath B is continuously drained by the cold secondary reservoir via $\mathcal{L}_{reset}$. In steady state, the flow of entropy is $S \to B \to$ reservoir, and the system is driven toward its ground state.*

IV. SPECTRAL ANALYSIS OF THE EXPANDER LINDBLADIAN

A. Structure of the Lindbladian Spectrum

The Lindbladian superoperator $\mathcal{L}_{full}$ acts on $\mathcal{B}(\mathcal{H})$ and is a linear map on an operator space. Its spectrum $\mathrm{spec}(\mathcal{L}_{full}) \subset \mathbb{C}$ consists of eigenvalues λ_k with $\mathrm{Re}(\lambda_k) \leq 0$, with $\lambda_0 = 0$ corresponding to the unique steady state. The spectral gap is $\Delta_{\mathcal{L}} = -\max_{k>0} \mathrm{Re}(\lambda_k) > 0$.

We decompose the Lindbladian into a coherent part and an incoherent part:

$$\mathcal{L}_{full} = -i\,\mathrm{ad}_{H_0} + \sum_{(i,j)\in E} \mathcal{D}_{L_{ij}} + \mathcal{L}_{reset}, \tag{9}$$

where $\mathrm{ad}_{H_0}[\rho] = [H_0, \rho]$. The coherent part is anti-Hermitian on the Hilbert-Schmidt space and does not affect the real parts of the eigenvalues of $\mathcal{L}_{full}$; all spectral gap bounds below concern the dissipative parts only.

B. Eigenvalue Bound from Expander Mixing

Lemma 1 (Expander Spectral Gap of the Dissipative Part)**.** *Let $\mathcal{L}_{diss} = \sum_{(i,j)\in E} \mathcal{D}_{L_{ij}}$ denote the purely dissipative part of the jump operator Lindbladian (without the reset). Let $\mathcal{L}_{diss}^{(exc)}$ be its restriction to the single-excitation sector, i.e., to density matrices supported on the subspace spanned by $\{|e_i\rangle\}$ for $i \in S \cup B$, where $|e_i\rangle$ is the state with a single excitation at site i. Then the spectral gap of $\mathcal{L}_{diss}^{(exc)}$ satisfies:*

$$\Delta_{\mathcal{L}_{diss}^{(exc)}} \geq \gamma \Delta_G, \tag{10}$$

where $\Delta_G = d - \lambda_2(A)$ is the spectral gap of the expander adjacency matrix A.

Proof. In the single-excitation sector, the action of $\mathcal{D}_{L_{ij}}$ on the amplitude vector $|\psi\rangle = \sum_k \alpha_k |e_k\rangle$ is that of a continuous-time random walk on the graph G: moving the excitation from site i to site j and vice versa with rate γ per edge. The generator of this random walk restricted to the single-excitation sector is $-\gamma L_{rw}$, where $L_{rw} = dI - A$ is the graph Laplacian, $D = dI$ is the degree matrix, and A is the adjacency matrix. The eigenvalues of $-\gamma L_{rw}$ are $-\gamma(d - \lambda_k(A))$ for $k = 0, 1, \ldots, |V| - 1$. The spectral gap (the smallest nonzero eigenvalue of γL_{rw}) is therefore:

$$\Delta_{\mathcal{L}_{diss}^{(exc)}} = \gamma(d - \lambda_2(A)) = \gamma \Delta_G, \tag{11}$$

which is exactly the stated bound. Note that the coupling rate γ in Eq. (4) is defined per edge, so no additional normalization by d is required. □

Corollary 2. *For a d-regular Ramanujan expander graph with $|V| = n + N$ vertices, the mixing time in the single-excitation sector satisfies:*

$$\tau_{mix}^{(1)} \leq \frac{1}{\gamma \Delta_G} \log\left(\frac{n+N}{\epsilon}\right) = \mathcal{O}\left(\frac{1}{\gamma} \log \frac{n+N}{\epsilon}\right), \tag{12}$$

for any target precision $\epsilon > 0$. In the canonical case $N = n$, this reads $\mathcal{O}(\gamma^{-1} \log(N/\epsilon))$, which is exponentially faster than the $\mathcal{O}(N^2)$ mixing time of a one-dimensional diffusive chain.

C. Full Lindbladian Gap with Reset

Adding the reset dissipators breaks the conservation of excitation number, as bath qubits are continuously returned to the vacuum. This is crucial for cooling; without the reset, the Lindbladian $\mathcal{L}$ (without $\mathcal{L}_{reset}$) would have a continuum of steady states corresponding to different excitation numbers.

Estimate 3 (Effective Gap of the Full Lindbladian)**.** *Let $\kappa > 0$ be the reset rate. While the exact spectral gap of a sum of non-commuting Lindbladians is difficult to bound rigorously for all parameter regimes, asymptotic physical arguments yield the effective relaxation rate:*

$$\Delta_{\mathcal{L}_{full}} \sim \min(\kappa,\, \gamma \Delta_G)\,. \tag{13}$$

Physical Justification. The coherent part $-i\,\mathrm{ad}_{H_0}$ is anti-Hermitian and does not reduce the real parts of eigenvalues. We therefore focus on $\mathcal{L}_{diss} + \mathcal{L}_{reset}$.

Both $\mathcal{L}_{diss}$ and $\mathcal{L}_{reset}$ are negative semidefinite superoperators, so $\mathcal{L}_{full}$ is also negative semidefinite, confirming the existence of a non-negative spectral gap. The scaling behavior is established by examining the asymptotic parameter regimes:

In regime (i), $\kappa \gg \gamma\Delta_G$, the bath is rapidly reset and the bottleneck is the expander dynamics. Adiabatic elimination of the bath degrees of freedom [16] yields an effective system Lindbladian whose gap scales as $\Theta(\gamma\Delta_G)$. This is the regime that the rest of the paper will assume: $\kappa \gg \gamma$.

In regime (ii), $\kappa \ll \gamma\Delta_G$, the bath excitation transport equilibrates rapidly and the bottleneck is the slow reset. The relevant relaxation rate is set by κ, the rate at which entropy is expelled from the bath.

While a strictly rigorous, uniform lower bound for the sum of non-commuting dissipative processes would require system-specific quantum Poincaré and logarithmic Sobolev inequalities [10], this asymptotic matching accurately captures the dynamical bottlenecks of the architecture. □

D. Steady State and Its Structure

Proposition 4 (Steady State Characterization)**.** *Under the dynamics of $\mathcal{L}_{full}$ in the regime $\kappa \gg \gamma$, the unique steady state ρ_{ss} satisfies:*

$$\rho_{ss} \approx \frac{e^{-\beta_{eff} H_S}}{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\otimes N}, \tag{14}$$

where $\mathcal{Z} = \mathrm{Tr}(e^{-\beta_{eff} H_S})$ is the partition function. The effective inverse temperature is determined by detailed balance between the excitation exchange rate γ and the bath reset rate κ. In steady state, detailed balance gives $p_{exc}/(1-p_{exc}) = \gamma/\kappa$ in the $\kappa \gg \gamma$ limit; equating this ratio to the Boltzmann factor $e^{-\beta_{eff}\omega_0}$ yields the exact relation:

$$\beta_{eff} = \frac{1}{\omega_0} \ln\left(\frac{\kappa}{\gamma}\right). \tag{15}$$

Ground state population approaches unity as $\kappa/\gamma \to \infty$.

Proof sketch. In the regime $\kappa \gg \gamma$, the bath is always close to its vacuum. Adiabatic elimination of the bath degrees of freedom [16] yields an effective Lindbladian on $\mathcal{H}_S$. For

a bath qubit j connected to system qubit i via a swap-type interaction of strength γ, the rate of excitation transfer from system to bath is $\gamma(1-p_{exc})$ and from bath to system is γp_{exc}, while the bath qubit is reset at rate κ. In steady state and in the $\kappa \gg \gamma$ limit, $\gamma(1-p_{exc}) = (\kappa+\gamma)p_{exc}$, giving $p_{exc} = \gamma/(\kappa+\gamma)$. Rearranging yields $p_{exc}/(1-p_{exc}) = \gamma/\kappa$. For a two-level system the Boltzmann ratio of populations is $p_{exc}/(1-p_{exc}) = e^{-\beta_{eff}\omega_0}$; Eq. (15) follows directly. The effective system Lindbladian satisfies detailed balance with respect to the Gibbs state $e^{-\beta_{eff}H_S}/\mathcal{Z}$, establishing it as the unique steady state. □

V. THE COMPLEXITY AND PURITY BOTTLENECKS

Despite the elegant mathematical convergence of the expander Lindbladian, mapping this to a physical architecture that solves hard problems reveals strict limitations. In this section we state and prove the main bottleneck results.

A. The Exponential Purity Requirement

Theorem 5 (Purity Bottleneck)**.** *Let H_S be a hard Hamiltonian in the sense of Definition 1, with exponentially small spectral gap $\Delta E \leq \Delta_0 \cdot 2^{-\alpha n}$. To prepare the system in the ground state $|E_0\rangle$ with fidelity $\mathcal{F} = \langle E_0|\rho_S|E_0\rangle \geq 1-\epsilon$ for any constant $\epsilon < 1/2$, the effective inverse temperature of the bath must satisfy:*

$$\beta_{eff} \geq \frac{\ln(1/\epsilon - 1) + \ln \dim \mathcal{H}_S}{\Delta E} \geq \Omega\left(\frac{n \cdot 2^{\alpha n}}{\Delta_0}\right). \tag{16}$$

Proof. The system steady state is the Gibbs state $\rho_{ss}^S = e^{-\beta_{eff}H_S}/\mathcal{Z}$. The ground state population is:

$$\langle E_0|\rho_{ss}^S|E_0\rangle = \frac{e^{-\beta_{eff}E_0}}{\mathcal{Z}} = \frac{1}{1+\sum_{k>0} e^{-\beta_{eff}(E_k-E_0)}}. \tag{17}$$

For $\mathcal{F} \geq 1-\epsilon$, we need $\sum_{k>0} e^{-\beta_{eff}\Delta E_k} \leq \epsilon/(1-\epsilon)$, where $\Delta E_k = E_k - E_0 \geq \Delta E$ for all $k > 0$. Since there are at most $D = 2^n$ excited states, the sufficient condition is $De^{-\beta_{eff}\Delta E} \leq \epsilon/(1-\epsilon)$. Rearranging:

$$\beta_{eff} \geq \frac{\ln D + \ln((1-\epsilon)/\epsilon)}{\Delta E} \geq \frac{n \ln 2}{\Delta_0 \cdot 2^{-\alpha n}} = \frac{n \ln 2 \cdot 2^{\alpha n}}{\Delta_0}. \tag{18}$$

This establishes the exponential lower bound on β_{eff}. □

Corollary 6 (Purity of Bath Qubits)**.** *Under the expander cooling protocol, the bath purity requirement translates to a super-exponential constraint on the ratio κ/γ. From Proposition 4, the effective inverse temperature is $\beta_{eff} = \omega_0^{-1}\ln(\kappa/\gamma)$, and the steady-state defect probability of each bath qubit is $p = \gamma/(\kappa+\gamma)$. Combining with Theorem 5, the requirement $\beta_{eff} \geq n\ln 2 \cdot 2^{\alpha n}/\Delta_0$ translates to:*

$$\frac{\kappa}{\gamma} \geq \exp\left(\frac{\omega_0 n \ln 2 \cdot 2^{\alpha n}}{\Delta_0}\right), \tag{19}$$

which is super-exponential in n. Equivalently, the defect probability must satisfy:

$$p \leq \exp\left(-\frac{\omega_0 n \ln 2 \cdot 2^{\alpha n}}{\Delta_0}\right), \tag{20}$$

i.e., the defect rate is super-exponentially suppressed. The purity of the bath state $\rho_B = \bigotimes_{k=1}^{N} \rho_k$, where $\rho_k = (1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|$, satisfies $P = \mathrm{Tr}(\rho_B^2) = (1-2p(1-p))^N$. For fixed N, $P \to 1$ as $p \to 0$; but achieving the required p of Eq. (20) demands a super-exponentially pure initial reservoir state.

Remark 2 (Physical Interpretation)**.** *Corollary 6 can be understood in terms of Landauer's principle. Driving the system into a state of low entropy (the ground state) requires exporting an entropy $\Delta S \approx n\ln 2$ into the bath. By Landauer's principle, this costs at least $k_B T_{eff}\Delta S = k_B T_{eff} n \ln 2$ of free energy, and the bath must be cool enough to absorb this entropy without being significantly excited. With $k_B T_{eff} = \omega_0/\ln(\kappa/\gamma)$, the condition $k_B T_{eff} \ll \Delta E \sim \Delta_0 \cdot 2^{-\alpha n}$ requires $\ln(\kappa/\gamma) \gg \omega_0/(\Delta_0 \cdot 2^{-\alpha n}) = \omega_0 \cdot 2^{\alpha n}/\Delta_0$, consistent with Eq. (19). This establishes that the computational hardness is encoded directly in the thermodynamic cost of preparing the bath.*

B. Hardness of Non-Local Simulation

Theorem 7 (Simulation Overhead)**.** *Compiling the expander Lindbladian $\mathcal{L}_{full}$ to a native-gate quantum circuit on a 2D superconducting qubit architecture (with nearest-neighbor connectivity) requires a circuit of depth $\mathcal{D}_{circ}$ satisfying:*

$$\mathcal{D}_{circ} \geq \Omega\left(\sqrt{n+N} \cdot \log(n+N)\right). \tag{21}$$

In the canonical case $N = n$, this gives $\mathcal{D}_{circ} \geq \Omega(\sqrt{N}\log N)$.

Proof sketch. The argument proceeds in two steps.

First, implementing a single jump operator L_{ij} (Eq. (4)) between two qubits i and j on a 2D grid of $n+N$ qubits requires a chain of SWAP gates of length $\Omega(\sqrt{n+N})$ to bring i and j into proximity. This lower bound is tight: the diameter of a $\sqrt{n+N} \times \sqrt{n+N}$ grid is $\Theta(\sqrt{n+N})$, so any pair of qubits that the expander connects at long range requires at least $\Omega(\sqrt{n+N})$ SWAP gates. A more elementary argument reaches the same bound via the information-theoretic observation that communicating one bit across a 2D grid of linear dimension ℓ requires $\Omega(\ell)$ steps.

Second, the expander graph has $|E| = d(n+N)/2 = \Theta(n+N)$ edges. Even with full parallelism, the edges of a d-regular expander cannot all be compiled simultaneously without conflicts; by Vizing's theorem, the chromatic index satisfies $\chi'(G) \in \{d, d+1\}$, so at least $d = \Theta(1)$ rounds of edge-disjoint SWAP chains are needed. Each round has depth $\Omega(\sqrt{n+N})$, giving a total depth per Trotter step of $\Theta(\sqrt{n+N})$.

To achieve the mixing time $\tau_{mix} = \mathcal{O}(\log(n+N)/\Delta_{\mathcal{L}_{full}})$, the number of Trotter steps required is $\tau_{mix}/\delta t = \mathcal{O}(\gamma d \log(n+N)/\Delta_{\mathcal{L}_{full}})$, which equals $\mathcal{O}(\log(n+N))$ using the asymptotic scaling of Estimate 3 and taking $\delta t \sim 1/(\gamma d)$. The total circuit depth is therefore $\Omega(\sqrt{n+N} \cdot \log(n+N))$. □

Corollary 8 (Partial Negation of Expander Speedup)**.** *On a 2D superconducting architecture, the total circuit depth to achieve the expander mixing time is $\Omega(\sqrt{N}\log N)$ (taking $N = n$), while the diffusive Lindbladian on the same 2D grid has native $\mathcal{O}(1)$-depth gates and requires $\mathcal{O}(N)$ time steps for mixing, giving total depth $\mathcal{O}(N)$. Both estimates assume each time step maps to $\mathcal{O}(1)$ native gate layers for the diffusive case, and $\mathcal{O}(\sqrt{N})$ SWAP layers per Trotter step for the expander. Under these assumptions:*

- *Expander cooling: total circuit depth $\Omega(\sqrt{N}\log N)$.*
- *Diffusive cooling (2D lattice): total circuit depth $\mathcal{O}(N)$.*

For large N, the expander is faster in total depth (since $\sqrt{N}\log N \ll N$), but the speedup is sub-polynomial rather than the naive exponential factor suggested by comparing mixing times alone. Neither protocol achieves polylogarithmic total circuit depth, and neither can efficiently solve hard problems.

C. Non-Locality of Energy-Selective Jump Operators

The jump operators L_{ij} defined in Eq. (4) are two-body operators that conserve the total excitation number. The spectral analysis of Section IV applies directly to these operators. However, for the cooling protocol to be thermodynamically selective (driving the system toward low-energy states rather than arbitrary dark states), one must in general engineer *energy-selective* jump operators that couple system modes with the correct energy difference to bath modes. The energy-selective version of L_{ij} involves local Bohr frequency projectors acting on the interaction neighborhood of spin i:

$$\tilde{L}_{ij}^{(\Delta E)} = \sqrt{\gamma}\, \Pi_{\Delta E}^{(i)} \cdot \sigma_-^{(i)} \sigma_+^{(j)}, \tag{22}$$

where $\Pi_{\Delta E}^{(i)}$ is the *local Bohr frequency projector* for spin i: it projects onto the subspace in which flipping spin i costs energy ΔE, which depends only on the spins in the interaction neighborhood of i under H_S. For a k-local H_S with interaction graph of maximum degree $\deg_{\max}$, the weight of $\Pi_{\Delta E}^{(i)}$ is bounded by $\deg_{\max} + 1 = \mathcal{O}(k)$. In contrast to a projector onto a *global* energy eigenspace of H_S - which would be an n-body operator - these local Bohr frequency projectors have support limited to the interaction neighborhood of i, and their implementation on local hardware requires $\mathcal{O}(2^{\deg_{\max}})$ elementary gates per projector. This represents an additional, qualitative source of overhead beyond the SWAP routing cost of Theorem 7: the jump operators required for physical cooling involve more than the basic two-body exchange operators of Eq. (4). For $k = \mathcal{O}(\log n)$ (as in many hard instances), the projector implementation cost is polynomial in n; for constant k, it adds a constant multiplicative overhead. In either case, the total circuit depth bound of Theorem 7 is a lower bound, and realistic protocols will exceed it due to the projector overhead. The steady-state analysis of Proposition 4 assumes that the energy-selective dissipation has been implemented; the simple operators of Eq. (4) do not by themselves produce a Gibbs state but rather a dark state that depends on the graph symmetries.

VI. COMPARISON WITH DIFFUSIVE COOLING

A. Mixing Times: A Quantitative Comparison

Table I summarizes the key parameters for three cooling protocols on N total qubits (taking $N = n$ for the expander row).

Protocol	Mixing Time	SWAP Cost per Step	Total Circuit Depth
1D Diffusive Chain	$\mathcal{O}(N^2)$	$\mathcal{O}(1)$	$\mathcal{O}(N^2)$
2D Diffusive Lattice	$\mathcal{O}(N)$	$\mathcal{O}(1)$	$\mathcal{O}(N)$
d-Regular Expander (2D hw.)	$\mathcal{O}(\log N)$	$\mathcal{O}(\sqrt{N})$	$\mathcal{O}(\sqrt{N}\log N)$

TABLE I. Comparison of cooling protocols for N total qubits on a 2D superconducting architecture (with $N = n$ for the expander). The SWAP cost is the gate count per Trotter step. The total circuit depth includes both the Trotter step count and SWAP cost per step, under the assumption that diffusive steps map to $\mathcal{O}(1)$ native gate layers and expander steps require $\mathcal{O}(\sqrt{N})$ SWAP layers each.

B. Spectral Gap Comparison

In the 1D diffusive model, the Lindbladian spectral gap scales as $\Delta_{\mathcal{L}}^{(1D)} \sim 1/N^2$ (diffusive dynamics). In the expander model, $\Delta_{\mathcal{L}}^{(exp)} \sim \gamma\Delta_G = \Theta(\gamma)$ (independent of N for a fixed-degree expander). This is the key advantage: the expander gap is *extensive* (size-independent), while the diffusive gap is *intensive* (shrinks with system size), giving an exponential improvement in mixing time.

However, as established in Theorem 7, this advantage is partially offset by the SWAP overhead. The *effective gap* for determining total circuit depth is not $\Delta_{\mathcal{L}}$ alone, but rather $\Delta_{\mathcal{L}}/(\text{gates per step})$: for the expander, this is $\Theta(\gamma/\sqrt{N})$; for the 2D diffusive lattice, it is $\Theta(1/N)$. The expander is still better by a factor of $\sqrt{N}$, but not by the exponential factor suggested by comparing mixing times in isolation.

VII. DISCUSSION

A. Potential Mitigations and Their Limits

1. *Quantum Error Correction for Bath Purity*

One might hope to use quantum error correction (QEC) to maintain the bath in a highly pure state despite imperfections. However, conventional QEC itself requires the ancilla qubits used for syndrome measurement to be in a pure state, a circular dependency. More formally, the overhead of QEC to maintain p-level purity on N physical qubits requires $\mathcal{O}(\log(1/p))$ layers of error correction, each of which consumes fresh ancilla qubits. For $p = \exp(-\Omega(n \cdot 2^{\alpha n}))$ as required by Eq. (20), the number of error correction layers needed is itself $\Omega(n \cdot 2^{\alpha n})$, and the total ancilla cost is exponential.

This still leaves one possible evasion - if we can re-encode a hard Hamiltonian such that required p has $p = \exp(-\Omega(n^k))$, then the number of error correction layers necessary is polynomial. In practice, for now, even this is too much.

Note that even if we only require polynomial inverse temperature β_{eff}, instead of exponential, this still requires significant purity on the bath - polynomial inverse temperature effectively corresponds to exponential purity. Therefore, β_{eff} needs to be reduced to constant to be practical for now.

2. *Quantum LDPC Codes and Embedding*

Recent advances in quantum low-density parity check (qLDPC) codes [17, 18] have shown that constant-rate, constant-distance quantum codes with logarithmic confinement can be constructed on graphs with expander structure. While qLDPC codes do provide a route to lower-overhead fault tolerance, the problem remains that the logical operations corresponding to the expander jump operators still involve non-local physical operations, and the logical-to-physical overhead does not reduce the circuit depth below $\Omega(\sqrt{N})$ on a 2D grid without changing the hardware topology.

3. Hardware with Native Long-Range Connectivity

If the underlying hardware natively supports the required long-range interactions (e.g., all-to-all connectivity as in ion trap systems [19], or the specific expander connectivity in a custom photonic chip), then the SWAP overhead of Theorem 7 is eliminated. In this case, the total circuit depth for expander cooling is $\mathcal{O}(\log N)$, recovering the exponential advantage in mixing time. However, the purity bottleneck of Theorem 5 and Corollary 6 remains: the hardware must still achieve a ratio κ/γ of the super-exponential magnitude given by Eq. (19), and this initialization is itself a hard thermodynamic problem.

B. Information-Theoretic Lower Bounds

The bottlenecks identified above can be understood through a unified information-theoretic lens. The task of cooling a system with n qubits from a maximally mixed state $I/2^n$ to its ground state $|E_0\rangle$ requires reducing the system entropy by $S = n \ln 2$ bits. By the data processing inequality, this entropy must have been exported into the environment. The environment (bath plus reservoir) must therefore have the capacity to absorb $n \ln 2$ bits of entropy while remaining in a state that can be recycled or reset. This requires the environment to have at least n bits of entropy reserve, which translates directly into the super-exponential purity requirement for the bath when the ground state has exponentially small Boltzmann weight at any polynomially achievable temperature.

This argument is essentially a quantum generalization of the Landauer bound and shows that no cooling protocol, regardless of its mixing topology, can circumvent the exponential purity requirement when the ground state has exponentially small Boltzmann weight at any polynomially achievable temperature.

VIII. CONCLUSION

An extensive analysis of an open quantum system cooling protocol based on non-local expander graph dynamics was presented in this paper, which can be summarized as follows.

1. **Mathematical construction:** We formally defined the expander Lindbladian with energy-conserving jump operators and continuous bath resupply (Sec. III), allowing

the bath size N to differ from the system size n, with $N = n$ as the canonical balanced case.

2. **Spectral analysis:** We proved that the spectral gap of the purely dissipative part in the single-excitation sector is exactly $\gamma\Delta_G$ (Lemma 1, and established the asymptotic scaling of the effective gap as $\sim \min(\kappa, \gamma\Delta_G)$ for the full Lindbladian (Estimate 3). We characterized the steady state as an exact Gibbs state at effective inverse temperature $\beta_{eff} = \omega_0^{-1} \ln(\kappa/\gamma)$ (Proposition 4), where this expression is asymptotically exact.

3. **Purity bottleneck:** We proved (Theorem 5 and Corollary 6) that cooling a hard Hamiltonian to its ground state requires a bath defect probability $p \leq \exp(-\Omega(n \cdot 2^{\alpha n}))$, a super-exponential constraint. This is stronger than naive estimates and follows from the asymptotically exact effective temperature formula. The computational hardness migrates from the system dynamics to the thermodynamic cost of the bath.

4. **Simulation overhead:** We showed (Theorem 7) that implementing the expander Lindbladian on a 2D superconducting architecture requires $\Omega(\sqrt{n+N}\log(n+N))$ circuit depth, reducing (but not eliminating) the mixing-time advantage. We also identified (Sec. V C) that thermodynamically selective cooling requires local Bohr frequency projectors whose weight is bounded by the maximum degree of the interaction graph - not the weight of global energy projectors, which would be n-body operators - constituting an additional source of overhead not captured by the mixing-time analysis alone.

5. **Comparison and discussion:** We quantitatively compared the expander approach to diffusive cooling (Sec. VI) and systematically analyzed potential mitigations (Sec. VII), concluding that none overcome the fundamental purity bottleneck.

While the expander topology provides a genuine improvement in *mixing time*, this cannot yet be converted into an efficient ground state preparation protocol for hard Hamiltonians - the bottleneck migrates from the dynamical mixing time to the thermodynamic preparation cost of the environment. Whether an environment exponentially close to a pure state can be prepared despite current practical difficulties is a task that future works should clarify.

DATA AVAILABILITY AND DECLARATION OF INTERESTS

The author(s) have no funding source to declare. Furthermore, there is no conflict of interests.

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